

Problem Statement & Proposed Solution

Executive assessment

Evidence Locker genuinely addresses the documentation and coordination problems faced by cyber-fraud victims and support teams. It provides a coherent path from incident capture to evidence organization, integrity recording, collaboration, and complaint-package generation.

The solution is strongest as a preparation and traceability layer. It reduces fragmented information and avoidable clarification cycles, but it cannot control external outcomes such as FIR registration, investigation decisions, fund recovery, or conviction.

Overall fit

STRONG

Primary role

ENABLER

External outcomes **LIMITED**

Purpose of this document

This document identifies the target problem, describes the proposed solution, maps capabilities to needs, and tests whether the current implementation is sufficient for the intended use case.

Scope note

The assessment is based on the current Evidence Locker repository implementation, including the React frontend, Express API, MongoDB models, evidence utilities, audit routes, and PDF generation utility.

1. Problem Statement

Cyber-fraud reporting is often a fragmented, high-stress process. A victim may need to reconstruct a timeline from payment records, messages, screenshots, emails, calls, social-media profiles, and device files. The information is then shared with helpers, banks, cyber cells, or police reviewers who need a clear, verifiable account.

Observed problem dimensions

- **Incomplete or inconsistent narratives**

Victims may omit key dates, identifiers, amounts, recipients, platforms, or category-specific facts because the reporting process is not sufficiently guided.

- **Evidence separated from explanation**

A screenshot, recording, document, or transaction record may be available without a clear source, capture context, relevance statement, or connection to the incident timeline.

- **Unclear integrity and handling history**

Reviewers may not know whether a file changed after collection, when it was uploaded, who handled it, or what actions were performed.

- **Slow and ambiguous collaboration**

Support teams and reviewers may exchange incomplete files and repeated clarification requests without a shared status model or clear ownership.

- **High operational and privacy risk**

Complaint records contain personal and sensitive information that requires authentication, access control, secure storage, and auditability.

Why the problem matters

- Missing facts can delay review and create repeated requests for clarification.
- Poorly described evidence is harder for a reviewer to interpret and connect to the complaint.
- Untracked handling reduces confidence in the evidence record and the completeness of the handoff.
- Sensitive personal and financial information can be exposed if access is not controlled.

2. Proposed Solution

Evidence Locker is a secure, role-aware complaint support platform that converts a cyber-fraud incident into a structured record and evidence package. It combines guided data capture, evidence intake, metadata, integrity hashing, custody records, status workflows, audit logging, and PDF export.

Core solution components

Guided complaint capture

Required fields and fraud-specific validation for UPI fraud, phishing, social-media harassment, loan-app fraud, and other incidents.

Evidence intake and storage

Controlled uploads for common media and document formats, linked to the relevant complaint and stored through the evidence service.

Metadata and context

Automatic extraction where available, combined with manual fields for source, device, transaction, platform, category, and limitations.

Integrity and custody

SHA-256 file hashes and custody-log events make post-upload integrity and handling history visible.

Role-based workflow

Victim, support, police, and administrator roles with controlled actions, status transitions, handoffs, and clarification paths.

Complaint package generation

PDF output containing complaint facts, structured details, evidence references, hashes, custody table, and a Section 65B draft template.

Proposed value

The platform does not promise a legal outcome; it improves the quality, consistency, traceability, and readiness of the material submitted for official review.

3. Problem-to-Solution Assessment

The current implementation addresses the central operational problems directly. The fit is strongest where the problem is information quality, evidence organization, access control, and workflow visibility.

Problem	Current response	Assessment
Incomplete narratives	Required complaint fields and fraud-specific validation.	Strong
Evidence without context	Metadata extraction plus manual evidence metadata and category fields.	Strong
Unclear integrity	SHA-256 hashing and visible evidence records.	Meaningful, with limits
Untracked handling	Custody logs and audit events for evidence and workflow actions.	Strong
Slow collaboration	Roles, statuses, handoff details, and clarification states.	Strong
Sensitive information risk	OTP/JWT auth, RBAC, encrypted contact fields, headers, and upload validation.	Strong foundation
Official outcome uncertainty	The platform documents and prepares; authorities decide.	Not controllable

Assessment conclusion

- The proposed solution is not merely descriptive: the repository contains implemented routes, models, validation, storage utilities, audit support, and PDF generation that correspond to the identified needs.
- The solution genuinely reduces documentation and coordination friction, but its impact depends on user adoption, data quality, secure deployment, and cooperation from external institutions.

4. Boundaries, Risks, and Success Measures

What the solution cannot guarantee

- FIR registration, police acceptance, investigation progress, fund recovery, or conviction.
- That a file was original before upload. A platform hash primarily supports integrity from hashing and storage onward.
- Complete metadata for every file. Source files and devices may not expose all relevant metadata.
- Legal sufficiency of a generated packet without review by the appropriate authority or legal professional.

How to measure whether it works

- Completion rate for mandatory and fraud-specific complaint fields.
- Percentage of evidence items with hashes, usable metadata, and clear relevance descriptions.
- Time from complaint creation to package-ready status.
- Number of clarification cycles caused by missing facts or evidence context.
- Audit completeness for authentication, evidence handling, and complaint handoffs.
- User and reviewer feedback on clarity, trust, and ease of submission preparation.

Final judgment

Yes. Evidence Locker genuinely addresses the identified documentation, evidence-context, integrity, collaboration, and privacy problems. It should be positioned as an enabling layer alongside—not instead of—official cyber-fraud reporting and investigation processes.

5. AI Chatbot Assistance

Evidence Locker also includes an authenticated AI chatbot that helps users understand digital incidents, organize evidence, build timelines, and prepare factual incident reports. The chatbot is an assistance layer inside the complaint workflow, not a replacement for a police officer, investigator, forensic examiner, lawyer, or emergency service.

How the AI chatbot addresses the problem

It provides calm, simple guidance; asks relevant follow-up questions; helps structure a timeline; and can use authorized evidence metadata for the linked incident without receiving the underlying files.

- Guidance during stressful reporting: explains what information is useful and what documentation steps to take next.
- Structured report support: uses only information supplied in the conversation and marks missing fields as “Not provided.”
- Evidence-aware assistance: can reference authorized metadata such as filenames, hashes, capture times, upload times, and descriptions when an incident is linked.
- Safe handling rules: does not ask for passwords, OTPs, private keys, or API keys, and does not fabricate, alter, authenticate, or claim evidence proves an allegation.
- Technical implementation: authenticated `/api/chat` route, configurable Gemini model, bounded conversation history, timeout handling, and explicit error responses when the AI service is unavailable.

Assessment of the AI component

Genuine contribution

Reduces uncertainty, improves question completeness, and helps users convert unstructured experiences into factual documentation.

Important boundary

AI output must be checked by the user or an authorized reviewer. It is not legal advice, forensic certification, or independent proof of an allegation.

Conclusion

The chatbot strengthens the proposed solution by making the platform more accessible to first-time reporters and support workers. Its value is highest when it helps users ask better questions, preserve facts accurately, and identify missing documentation while keeping human and official review in control.